

PHYS 143 – Mechanics and Thermal Physics

Thermal Properties of Matter

How do pressure, volume and temperature describe a gas, what happens at a change of phase, and what do molecules do?

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Reading: Young and Freedman, *University Physics* (15th ed.), Ch. 18 (§18.1, §18.3, §18.5, §18.6) and §17.5

By the end of this week, you should be able to:

1. Use the ideal-gas equation and the gas laws to calculate and compare the states of a gas.
2. Read pV and phase diagrams, and calculate the heat in a phase change (latent heat), including calorimetry.
3. State the assumptions of the kinetic theory and explain pressure and temperature in terms of molecular motion.
4. Calculate the average kinetic energy and the rms speed of molecules, and select the right model in an unfamiliar problem.

The detailed outcomes are in the checklist at the end of the lecture.

Three contacts:

1 DESCRIBE

How is the state of a gas described?

2 REPRESENT

What do the diagrams show, and what happens at a change of phase?

3 EXPLAIN

What do molecules do, and what do pressure and temperature mean?

Where we start: a five-minute check

Not marked. Answer individually, then we review.

1. Convert 25°C to kelvin and 2.0 atm to pascals ($1\text{ atm} = 1.013 \times 10^5\text{ Pa}$).
2. A force of 50 N acts on an area of 0.020 m^2 . Find the pressure.
3. A ball of mass 0.10 kg hits a wall at 20 m/s and bounces back at the same speed. Find the change in its momentum.
4. Find the kinetic energy of a 2.0 kg mass moving at 3.0 m/s .
5. If p is proportional to $1/V$, what happens to p when V is halved?
6. How much heat raises 0.50 kg of water by 20 K ($c = 4186\text{ J}/(\text{kg K})$)?

1. 298.15 K;
 $2.0 \times 1.013 \times 10^5 = 2.0 \times 10^5 \text{ Pa}.$
2. $p = F/A = 50/0.020 = 2.5 \times 10^3 \text{ Pa}.$
3. $\Delta p = (-2.0) - (+2.0) = -4.0 \text{ kg m/s}$
(the magnitude is $2mv$).
4. $K = \frac{1}{2}(2.0)(3.0)^2 = 9.0 \text{ J}.$
5. p doubles.
6. $Q = mc\Delta T = (0.50)(4186)(20) = 4.2 \times 10^4 \text{ J}.$

Kelvin and $Q = mc\Delta T$ (Lecture 9), pressure and momentum change (Lectures 3 and 7) and kinetic energy (Lectures 4 and 5) are reused. Lecture 9 ended with a question: what happens when a body reaches a melting or boiling point? That is answered in Contact 2.

Contact 1: Describe

How is the state of a gas described?

Poll

You close the nozzle of a bicycle pump with a finger and push the handle in quickly. The air inside becomes harder to compress, and the barrel becomes warm. **Which of the quantities pressure, volume, temperature and amount of air changed?**

Predict first

Sketch the barrel before and after. Which quantities change, and which stays fixed? Does the pressure rise by the same factor as the volume falls?

Poll

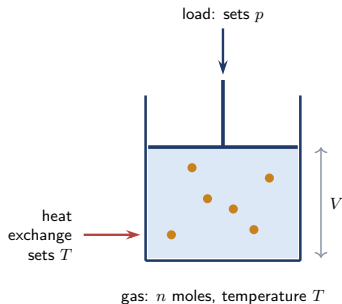
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Predict first

Sketch the barrel before and after. Which quantities change, and which stays fixed? Does the pressure rise by the same factor as the volume falls?

p , V and T all change; the amount of air (the number of molecules) stays fixed because the nozzle is closed. One equation should connect them. We will build it from experiment.

State variables and the equation of state



- The **state variables** of a gas: pressure p , volume V , temperature T and amount of substance n (moles).
- An **equation of state** relates them for a given material: fix three, and the fourth follows.
- T and p are intensive (independent of the amount); V and n are extensive.

Physical meaning

Large-scale measurements (p, V, T, n) describe the state of the gas without following any molecule. Lecture 9 gave the meaning of T ; this lecture connects p and T to molecules.

Rule: amount of substance

$$N = nN_A \quad n = \frac{m}{M} \quad m_0 = \frac{M}{N_A}$$

$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$; M : molar mass (kg/mol); m_0 : mass of one molecule.

Examples of molar mass:

nitrogen N_2 : 28.0 g/mol; carbon dioxide CO_2 : 44.0 g/mol; helium (monatomic): 4.00 g/mol.

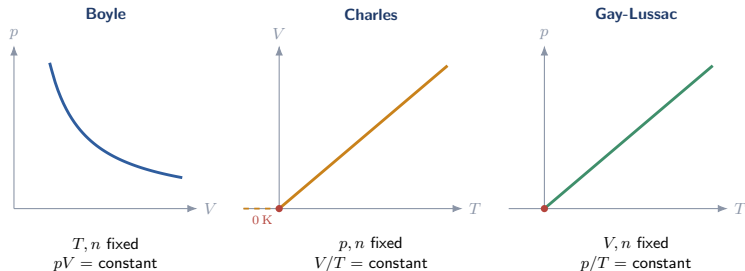
$$m_0(\text{N}_2) = \frac{28.0 \times 10^{-3}}{6.022 \times 10^{23}} = 4.65 \times 10^{-26} \text{ kg};$$

$$m_0(\text{CO}_2) = 7.31 \times 10^{-26} \text{ kg}.$$

Common error

Equal *masses* of two gases do not contain equal numbers of molecules. Equal *moles* do. 10 g of hydrogen has many more molecules than 10 g of oxygen.

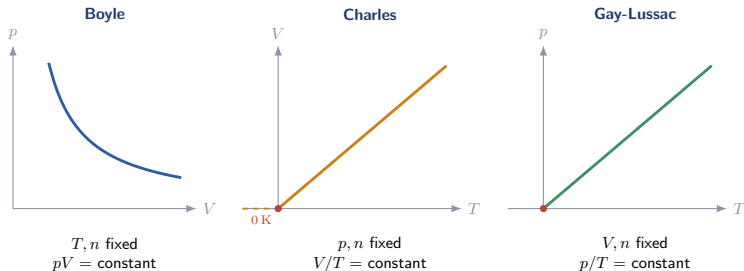
Three experiments with a fixed amount of gas



Predict first

Charles: a line through the origin only if T is in kelvin. If V is 2.0 L at 300 K and p is fixed, what is V at 600 K? And at 600°C?

Three experiments with a fixed amount of gas



Predict first

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At 600 K: 4.0 L. At 600°C = 873 K: 5.8 L, *not* 4.0 L. Doubling the Celsius number does not double the volume.

Rule: ideal-gas equation

$$pV = nRT = NkT$$

$R = 8.314 \text{ J}/(\text{mol K})$, $k = R/N_A = 1.381 \times 10^{-23} \text{ J/K}$. p absolute pressure; T in **kelvin**.

Physical meaning

Boyle, Charles and Gay-Lussac are this one equation with one variable held fixed: pV constant at fixed T ; V/T constant at fixed p ; p/T constant at fixed V .

Model validity

The ideal gas is a model: it works for *low density*, and temperatures well above condensation. Real gases deviate at high pressure and near the liquid state (Contact 2).

TOOLKIT 1 – State variables and units

1. **LIST** what is given: p , V , T , and either n , m or N . Which one is unknown?
2. **TEMPERATURE** in kelvin: $T = T_C + 273.15$ (never $^{\circ}\text{C}$ in a gas law).
3. **PRESSURE** in pascals, *absolute*: $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$; a gauge reads

$$p_{\text{gauge}} = p - p_{\text{atm}}.$$

4. **VOLUME** in m^3 ($1 \text{ L} = 10^{-3} \text{ m}^3$); **AMOUNT** in moles, with M in kg/mol .

WATCH OUT: using $^{\circ}\text{C}$, using a gauge pressure, or M in g/mol with R in $\text{J}/(\text{mol K})$.

CHECK: units: $\text{Pa m}^3 = \text{J}$, so pV and nRT are both energies.

TOOLKIT 2 – The ideal-gas equation

1. **IDENTIFY** one state of the gas: three of p , V , T , n known.
2. **CHOOSE the form:** $pV = nRT$ (moles) or $pV = NkT$ (molecules); with $n = m/M$ it gives the density $\rho = pM/(RT)$.
3. **SOLVE** for the unknown, then convert: $N = nN_A$, $m = nM$.
4. **PREDICT** first: at 1 atm and 0°C one mole occupies about 22.4 L.

WATCH OUT: use this form only for *one state*. For a *change* of state use the ratio method (Toolkit 3).

CHECK: is the number of moles of a sensible size for the container?

Question

A tank of volume 15.0 L holds helium at an absolute pressure of $3.0 \times 10^5 \text{ Pa}$ and a temperature of 22°C . Find (a) the number of moles, (b) the number of atoms, (c) the mass of helium.

Predict first

A mole of gas at 1 atm fills about 22 L. This tank is smaller but at about 3 atm: about 0.2, 2 or 20 moles?

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Predict first

A mole of gas at 1 atm fills about 22 L. This tank is smaller but at about 3 atm: about 0.2, 2 or 20 moles?

Set up: $V = 15.0 \times 10^{-3} \text{ m}^3$, $T = 295.15 \text{ K}$.

Execute: (a) $n = \frac{pV}{RT} = \frac{(3.0 \times 10^5)(15.0 \times 10^{-3})}{(8.314)(295.15)} = 1.83 \text{ mol}$. (b)

$N = nN_A = 1.10 \times 10^{24}$. (c) $m = nM = (1.83)(4.00 \text{ g/mol}) = 7.3 \text{ g}$.

Evaluate: about 2 moles, as predicted; 7.3 g of helium is a plausible mass for a small tank. ✓

The ratio method for a change of state

Rule: fixed amount of gas

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2} \quad (n \text{ constant}).$$

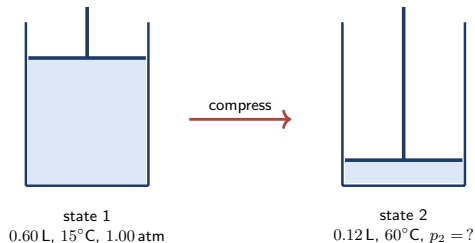
Same units on both sides; T in kelvin; p absolute.

✂ TOOLKIT 3 – The ratio method for a change of state

1. **STATES:** write the numbers for state 1 and state 2; mark the unknown.
2. **WHAT IS CONSTANT?** T (Boyle), p (Charles), or V (Gay-Lussac): cancel it.
3. **SOLVE** $p_2 = p_1 \frac{V_1}{V_2} \frac{T_2}{T_1}$ (or for the other unknown). Units of p and V cancel in the ratio.

WATCH OUT: n must not change (no leaks); T in kelvin. **CHECK:** does each ratio push the answer the way you predicted?

Lecturer example: compressing air in a pump



Air at 1.00 atm and 15°C fills a closed pump barrel of volume 0.60 L. It is compressed quickly to 0.12 L, and its temperature rises to 60°C. Find the final pressure.

Predict first

The volume falls by a factor of 5. Is the pressure exactly 5 atm, more, or less?

Set up: nothing leaks, so n is fixed. $T_1 = 288.15\text{ K}$, $T_2 = 333.15\text{ K}$.

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Evaluate: more than 5 atm, as predicted: the compression raises the pressure by 5, and the warming adds a further 16%. Using $^{\circ}\text{C}$ instead of kelvin would give the absurd ratio $60/15 = 4$. ✓

Your turn: a weather balloon

Your turn

A weather balloon is released with 4.0 m^3 of helium at 1.00 atm and 15°C . At a certain height the pressure is 0.30 atm and the temperature is -33°C . Find the volume, assuming no helium escapes and the gas pressure equals the outside pressure.

Predict first

Sketch the balloon at the two heights. Lower pressure makes the gas expand, lower temperature makes it shrink: which effect is larger? About 4, 11 or 40 m^3 ?

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$$V_2 = V_1 \frac{p_1}{p_2} \frac{T_2}{T_1} = (4.0) \left(\frac{1.00}{0.30} \right) \left(\frac{240.15}{288.15} \right) = 11 \text{ m}^3.$$

Evaluate: the pressure effect (factor 3.3) dominates the cooling (factor 0.83), as predicted. ✓

Concept check: a changing gas

Choose, then explain

A fixed amount of ideal gas is at 27°C . Its pressure is halved and its volume is tripled. What is the new temperature?

A. 177°C B. 27°C C. 40.5°C D. -123°C E. -177°C

Predict first

Does the product pV go up or down? What does that say about T (before any arithmetic)?

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Predict first

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pV is multiplied by $\frac{1}{2} \times 3 = 1.5$, so T rises by the same factor:

$T_2 = 1.5 \times 300.15 = 450 \text{ K} = 177^{\circ}\text{C}$ (A). Answer C comes from multiplying the Celsius number: the common error.

Contact 1: what we can now do

State variables p, V, T, n



Equation of state: $pV = nRT = NkT$



Change of state: $\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$

You should be able to

- define the state variables and use n , N , M ;
- convert to kelvin, pascals (absolute), m^3 ;
- use $pV = nRT$ and the ratio method;
- say when the ideal-gas model is valid.

✓ Before Contact 2

How can all the states of a gas be shown on one picture, and what happens when the gas condenses? Read Young and Freedman, §18.1 and §18.6.

Contact 2: Represent

*What do the diagrams show, and
what happens at a change of phase?*

An opening question: all the states on one graph

Sketch first

A fixed amount of ideal gas can be at many values of p and V . On a graph with V along the horizontal axis and p upward, sketch the curve joining all states at one temperature T_1 . Then add the curve for a higher temperature T_2 .

Predict first

Does the curve touch an axis? Does the higher-temperature curve lie closer to the origin or farther from it?

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Predict first

Does the curve touch an axis? Does the higher-temperature curve lie closer to the origin or farther from it?

The curve $p = nRT/V$ is a hyperbola: it never touches an axis, and a higher temperature gives a curve farther from the origin. Each such curve is an **isotherm**.

Rule: comparing states on a pV diagram

$$pV = nRT \quad \Rightarrow \quad \text{for the same } n : T \propto pV$$

The product pV (height times width of the rectangle under the point) tells the temperature. Equal pV : the same isotherm, the same temperature.

- An isotherm is a path at constant T (Boyle's law).
- A vertical path is constant V ; a horizontal path is constant p .
- Only the ideal gas gives exact hyperbolas.

Common error

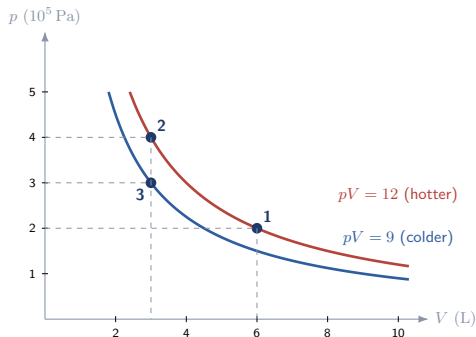
Comparing temperatures by looking only at p or only at V . A state with high p and small V can have the same temperature as one with low p and large V .

TOOLKIT 4 – Reading pV and phase diagrams

1. **AXES:** what is plotted (p against V , or p against T)? What is fixed (the amount)?
2. **pV DIAGRAM:** sketch the isotherm through each point; the farther from the origin, the higher the T . Compare pV products.
3. **PATHS:** horizontal = constant p ; vertical = constant V ; hyperbola = constant T .
4. **PHASE DIAGRAM:** read the region (solid, liquid, vapour); a path crossing a curve is a change of phase; on a curve two phases coexist.
5. **SPECIAL POINTS:** triple point (three phases), critical point (no distinction between liquid and vapour).

WATCH OUT: a phase diagram is p against T ; the pV diagram of a real gas has a liquid-vapour region. **CHECK:** does the path cross the curves you expect?

Lecturer example: which state is hottest?



0.50 mol of an ideal gas can be in three states:

1: $p = 2.0 \times 10^5 \text{ Pa}$, $V = 6.0 \text{ L}$;

2: $p = 4.0 \times 10^5 \text{ Pa}$, $V = 3.0 \text{ L}$;

3: $p = 3.0 \times 10^5 \text{ Pa}$, $V = 3.0 \text{ L}$.

Which state is at the highest temperature?
Find all three temperatures.

Predict first

Sketch the isotherms. Is one state clearly hottest, or are two tied?

Set up: $T = pV/(nR)$; $nR = (0.50)(8.314) = 4.157 \text{ J/K}$.

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Execute: $p_1 V_1 = (2.0 \times 10^5)(6.0 \times 10^{-3}) = 1200 \text{ J}$,

$p_2 V_2 = (4.0 \times 10^5)(3.0 \times 10^{-3}) = 1200 \text{ J}$, $p_3 V_3 = 900 \text{ J}$.

$T_1 = T_2 = 1200/4.157 = 289 \text{ K } (16^\circ\text{C})$; $T_3 = 900/4.157 = 217 \text{ K } (-57^\circ\text{C})$.

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Evaluate: states 1 and 2 are tied for the highest temperature and lie on the same isotherm; state 3 lies closer to the origin and is colder, as the sketch showed. A state with the largest p (state 2) is not necessarily the hottest alone. ✓

Your turn: a path on the diagram

Your turn

Starting from state 1 above (2.0×10^5 Pa, 6.0 L, 289 K), the gas is heated in a rigid container until its pressure doubles. Sketch the path on the pV diagram and find the final temperature.

Predict first

Is the path vertical or horizontal? Does it cross to a higher or lower isotherm?

Your turn: a path on the diagram

Your turn

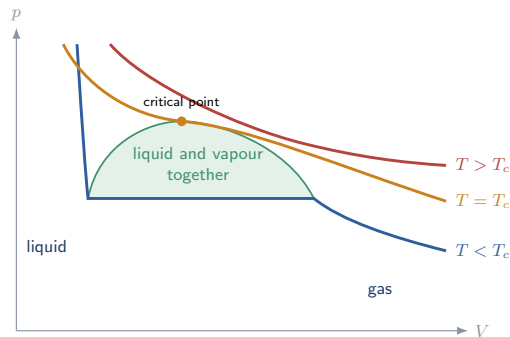
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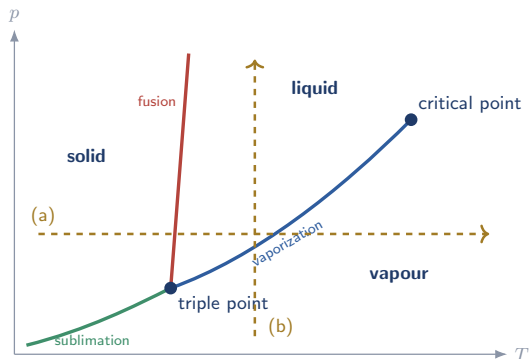
V is constant, so the path is vertical (upward). $T_2 = T_1 p_2/p_1 = 2 \times 289 = 577 \text{ K} = 304^\circ\text{C}$ (Gay-Lussac). **Evaluate:** the point is on a higher isotherm, as predicted; check $p_2 V_2/(nR) = (4.0 \times 10^5)(6.0 \times 10^{-3})/4.157 = 577 \text{ K}$. ✓

Real gases: the isotherms bend



- Above the critical temperature T_c the isotherms resemble the ideal gas; at high density or low T the ideal-gas equation fails.
- Below T_c , compression at constant T reaches a flat section: at constant pressure the vapour condenses, the volume shrinks, and liquid and vapour coexist.

The phase diagram: pressure against temperature



Rule: reading a phase diagram

Each point (T, p) has one phase, except on a *curve*, where two phases coexist in equilibrium. All three coexist only at the **triple point**. Past the **critical point** liquid and vapour cannot be distinguished.

Path (a): heating at constant p . Path (b): compression at constant T . Boiling and melting temperatures depend on pressure.

Question

The triple point of carbon dioxide is at 216.55 K and 5.17×10^5 Pa. A block of solid CO₂ (dry ice) is left in a room at 1 atm. Does it melt to a liquid? What would you have to do to obtain liquid CO₂?

Predict first

Sketch a horizontal line at 1 atm on the phase diagram. Is 1 atm above or below the triple-point pressure?

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Sketch a horizontal line at 1 atm on the phase diagram. Is 1 atm above or below the triple-point pressure?

1 atm = 1.013×10^5 Pa is about 0.2 of the triple-point pressure (5.10 atm), so the line passes below the triple point: it meets only the sublimation curve. The solid **sublimes** directly to vapour (at about -78°C). To obtain the liquid, the pressure must exceed 5.17×10^5 Pa.

Evaluate: this is why dry ice leaves no puddle. ✓

What happens at a change of phase?

Rule: latent heat

$$Q = \pm mL$$

+: heat enters (melting, boiling); -: heat leaves (freezing, condensing). L in J/kg.

Physical meaning

During the change the temperature stays *constant*: the energy goes into changing the arrangement of the molecules, not into raising their average kinetic energy.

Substance	L_f (J/kg)	L_v (J/kg)
water	3.34×10^5	2.256×10^6
ethanol	1.04×10^5	8.54×10^5
nitrogen	2.55×10^4	2.01×10^5

Young and Freedman, Table 17.4 (normal pressure). $L_v > L_f$: vaporization separates molecules completely.

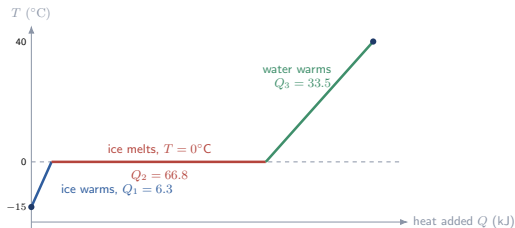
✂ TOOLKIT 5 – Heating curves and latent heat

1. **SKETCH** the T against Q curve: slanted lines in one phase, flat lines at a change of phase.
2. **LIST THE STAGES** from the initial to the final state (warm the solid, melt, warm the liquid, boil, warm the vapour).
3. **SLANTED STAGE:** $Q = mc\Delta T$ with c of *that* phase (Lecture 9).
4. **FLAT STAGE:** $Q = \pm mL$ at the transition temperature.
5. **ADD** the stages (all positive when heating); $Q = Pt$ turns energy into time.

WATCH OUT: using one c for all phases; forgetting a stage; using $mc\Delta T$ at a change of phase.

CHECK: does the largest bar on the sketch correspond to the largest Q ?

Lecturer example: from ice to warm water



How much heat is needed to take 0.200 kg of ice at -15°C to water at 40°C ?
($c_{\text{ice}} = 2100$, $c_{\text{water}} = 4186 \text{ J}/(\text{kg K})$,
 $L_f = 3.34 \times 10^5 \text{ J/kg.}$)

Predict first

Which stage takes the most heat: warming the ice, melting it, or warming the water? Why?

Execute: $Q_1 = mc_{\text{ice}}\Delta T = (0.200)(2100)(15) = 6.3 \times 10^3 \text{ J}.$

$$Q_2 = mL_f = (0.200)(3.34 \times 10^5) = 6.68 \times 10^4 \text{ J}.$$

$$Q_3 = mc_{\text{water}}\Delta T = (0.200)(4186)(40) = 3.35 \times 10^4 \text{ J}.$$

Execute: $Q_1 = mc_{\text{ice}}\Delta T = (0.200)(2100)(15) = 6.3 \times 10^3 \text{ J}.$

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$Q_3 = mc_{\text{water}}\Delta T = (0.200)(4186)(40) = 3.35 \times 10^4 \text{ J}.$

$Q = Q_1 + Q_2 + Q_3 = 1.07 \times 10^5 \text{ J}.$

Evaluate: melting takes the most (63%), as the flat line of the sketch is the longest, even though it involves no temperature change. Melting the ice needs more energy than warming the melt by 40 K. All terms are positive, as heat enters. ✓

Your turn: melting and boiling away

Your turn

- (a) A 1.5 kW heater supplies all its energy to 0.50 kg of ice already at 0°C . How long does it take to melt it?
- (b) A 2.0 kW kettle element boils away 1.00 kg of water that is already at 100°C . How long does that take?

Predict first

In Lecture 9 a kettle took 4.2 minutes to heat 1.5 kg of water from 20°C to 100°C . Is boiling away 1.00 kg faster, slower or about the same?

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$$(a) \ t = \frac{mL_f}{P} = \frac{(0.50)(3.34 \times 10^5)}{1500} = 1.1 \times 10^2 \text{ s} \approx 1.9 \text{ min.}$$

$$(b) \ t = \frac{mL_v}{P} = \frac{(1.00)(2.256 \times 10^6)}{2000} = 1.13 \times 10^3 \text{ s} \approx 19 \text{ min.}$$

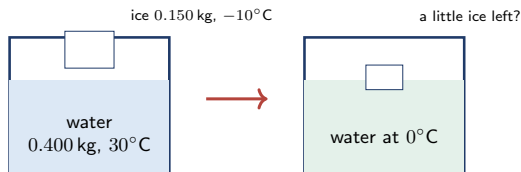
Evaluate: slower, as predicted: vaporization takes about five times the energy of heating the same water from 0 to 100°C (2.26×10^6 against 4.19×10^5 J). ✓

✂ TOOLKIT 6 – Calorimetry with a change of phase

1. **SYSTEM:** insulated; list every body, its phase and initial temperature.
2. **DECIDE THE FINAL STATE FIRST:** compute the heat the warmer body can give up in cooling to the transition temperature, and the heat the colder body needs to reach it and to complete its change of phase. Compare.
3. **WRITE** $\sum Q = 0$ with the terms that belong to *that* final state ($mc\Delta T$ and $\pm mL$).
4. **SOLVE** for the unknown (T_f , or the mass that changes phase).
5. **EVALUATE:** the result must be consistent with the assumed state.

WATCH OUT: assuming the final state; sign of the mL terms; forgetting the warming of the ice. **CHECK:** is $0 \leq m_{\text{melted}} \leq m_{\text{ice}}$?

Lecturer example: ice added to water



0.150 kg of ice at -10°C is dropped into 0.400 kg of water at 30°C in an insulated container. Find the final state.

Predict first

Sketch the heat flow. Will all the ice melt? Will the final temperature be above 0°C , exactly 0°C , or below?

Decide the final state: heat available from the water cooling to 0°C :

$0.400(4186)(30) = 5.02 \times 10^4 \text{ J}$. Heat needed to warm the ice to 0°C :

$0.150(2100)(10) = 3.2 \times 10^3 \text{ J}$, and to melt all of it: $0.150(3.34 \times 10^5) = 5.01 \times 10^4 \text{ J}$; total $5.33 \times 10^4 \text{ J}$.

Needed > available: not all the ice melts, and the final temperature is 0°C .

Decide the final state: heat available from the water cooling to 0°C :

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$0.150(2100)(10) = 3.2 \times 10^3 \text{ J}$, and to melt all of it: $0.150(3.34 \times 10^5) = 5.01 \times 10^4 \text{ J}$; total $5.33 \times 10^4 \text{ J}$.

Needed > available: not all the ice melts, and the final temperature is 0°C .

Execute: $m_{\text{melted}} = \frac{50232 - 3150}{3.34 \times 10^5} = 0.141 \text{ kg}$; ice left: $0.150 - 0.141 = 0.009 \text{ kg}$.

Evaluate: final: $T = 0^{\circ}\text{C}$ with 0.541 kg of water and 9 g of ice. The comparison was close (53.3 against 50.2 kJ), so the check mattered: guessing “all melts” would give a result below 0°C for the water. ✓

Your turn: steam into water

Your turn

0.020 kg of steam at 100°C is bubbled into 0.300 kg of water at 20°C in an insulated container. Find the final temperature ($L_v = 2.256 \times 10^6 \text{ J/kg}$).

Predict first

Decide the final state first: can 0.020 kg of steam heat the water all the way to 100°C? Will the final temperature be nearer 20°C or 100°C?

Your turn

0.020 kg of steam at 100°C is bubbled into 0.300 kg of water at 20°C in an insulated container. Find the final temperature ($L_v = 2.256 \times 10^6 \text{ J/kg}$).

Predict first

Decide the final state first: can 0.020 kg of steam heat the water all the way to 100°C? Will the final temperature be nearer 20°C or 100°C?

The steam condenses ($4.5 \times 10^4 \text{ J}$ released) but warming the water to 100°C would need $1.0 \times 10^5 \text{ J}$: so $T_f < 100^\circ\text{C}$ and all the steam condenses.

$$m_s L_v + m_s c(100 - T_f) = m_w c(T_f - 20) \Rightarrow T_f = \frac{45\,120 + 8372 + 25\,116}{1339.5} = 58.7^\circ\text{C}.$$

Evaluate: nearer to 20°C than to 100°C (with T_f between the two); heat released = 48.6 kJ = heat gained. ✓

Contact 2: what we can now do

pV diagram: $T \propto pV$; isotherms



phase diagram: regions, curves, triple point



change of phase: $Q = \pm mL$, heating curve



calorimetry: decide the final state, then $\sum Q = 0$

You should be able to

- compare states on a pV diagram;
- read a phase diagram and trace a path;
- calculate Q across a change of phase;
- solve a calorimetry problem with a phase change.

✓ Before Contact 3

$pV = NkT$ was found from experiment. Can molecules explain it? Read Young and Freedman, §18.3 and §18.5.

Contact 3: Explain

*What do molecules do, and what
do pressure and temperature mean?*

An opening question: where does pressure come from?

Poll

A gas in a sealed container presses on the walls. Nothing is pushing the gas against the walls. **What causes the pressure?** If every molecule moved twice as fast, by what factor would the pressure change?

Predict first

Sketch one molecule bouncing off a wall. Faster molecules hit harder, and they also hit more often. Does the pressure double, quadruple, or stay the same?

An opening question: where does pressure come from?

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Predict first

Sketch one molecule bouncing off a wall. Faster molecules hit harder, and they also hit more often. Does the pressure double, quadruple, or stay the same?

Pressure is the average force per unit area from the very many molecular collisions with the wall. Each collision is $2\times$ stronger and the collisions are $2\times$ as frequent: the pressure becomes $4\times$. We now make this precise.

The kinetic model of an ideal gas

Assumption

Where it is used

A very large number N of identical molecules of mass m

averages are meaningful

Random motion, with no preferred direction

$$(v_x^2)_{\text{av}} = (v_y^2)_{\text{av}} = (v_z^2)_{\text{av}}$$

Molecules are point-like: their volume is negligible

the whole container volume is free

No forces except in collisions

no potential energy; straight-line paths

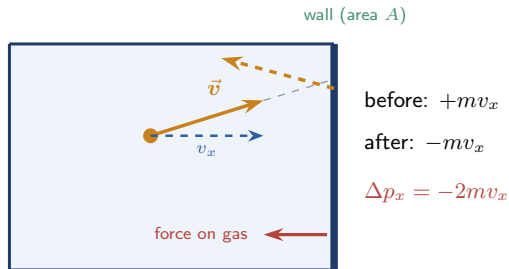
Collisions with the wall are elastic

the speed is unchanged, $v_x \rightarrow -v_x$

Physical meaning

Each assumption is a simplification. The model works best for low density, which is exactly where the ideal-gas equation works.

One molecule, one wall



A molecule in a cubical box of side L hits the right wall elastically.

- Change in the molecule's momentum:
 $\Delta p_x = -2mv_x$.
- It returns to the same wall after a time $2L/v_x$.
- Average force on the wall:

$$F = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L}.$$

The force on the wall is the rate of change of momentum (Lecture 7), with the sign reversed by Newton's third law.

From one molecule to a pressure

1. **Sum over N molecules:** $F = \frac{Nm(v_x^2)_{\text{av}}}{L}$.
2. **Three directions are equivalent:** $(v^2)_{\text{av}} = 3(v_x^2)_{\text{av}}$, so $(v_x^2)_{\text{av}} = \frac{1}{3}(v^2)_{\text{av}}$.
3. **Pressure is force per area** ($A = L^2$): $p = \frac{F}{A} = \frac{Nm(v^2)_{\text{av}}}{3L^3} = \frac{1}{3} \frac{N}{V} m(v^2)_{\text{av}}$.

Rule: pressure of an ideal gas

$$pV = \frac{1}{3}Nm(v^2)_{\text{av}} = \frac{2}{3}N \left[\frac{1}{2}m(v^2)_{\text{av}} \right]$$

The pressure grows with the number density and with the average translational kinetic energy per molecule.

Temperature has a molecular meaning

Rule: kinetic energy and temperature

Compare $pV = \frac{2}{3}N \left[\frac{1}{2}m(v^2)_{\text{av}} \right]$ with $pV = NkT$:

$$K = \frac{1}{2}m(v^2)_{\text{av}} = \frac{3}{2}kT \quad K_{\text{tot}} = \frac{3}{2}nRT$$

Physical meaning

Temperature measures the average translational kinetic energy per molecule. It depends only on T , not on p , V or the kind of gas.

Common errors

(1) “ T measures the total energy.” It is the average *per molecule*. (2) “Molecules stop at 0°C .” They stop only in the unreachable limit $T \rightarrow 0\text{ K}$.

Why root-mean-square?

Rule: rms speed

$$v_{\text{rms}} = \sqrt{(v^2)_{\text{av}}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

The mean of the *squares* is what appears in the kinetic energy, so v_{rms} follows directly; the average speed v_{av} is slightly smaller.

Check by hand

Four molecules have speeds 200, 400, 600 and 800 m/s. Find v_{av} and v_{rms} .

Why root-mean-square?

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Check by hand

Four molecules have speeds 200, 400, 600 and 800 m/s. Find v_{av} and v_{rms} .

$$v_{\text{av}} = 500 \text{ m/s}; \quad (v^2)_{\text{av}} = \frac{200^2 + 400^2 + 600^2 + 800^2}{4} = 3.0 \times 10^5 \text{ m}^2/\text{s}^2, \text{ so } v_{\text{rms}} = 548 \text{ m/s}.$$

$v_{\text{rms}} > v_{\text{av}}$: squaring weights the fast molecules more.

TOOLKIT 7 – Kinetic theory

1. **SKETCH** the molecules and the wall; state the assumptions that apply.
 2. **PER MOLECULE:** $K = \frac{3}{2}kT$ (energy in J); use $m_0 = M/N_A$ for the molecular mass.
 3. **SPEED:** $v_{\text{rms}} = \sqrt{3kT/m} = \sqrt{3RT/M}$ with M in **kg/mol**.
 4. **TOTAL:** $K_{\text{tot}} = NK = \frac{3}{2}nRT$ depends on the *amount*.
 5. **SCALING:** $v_{\text{rms}} \propto \sqrt{T}$ and $\propto 1/\sqrt{M}$; $K \propto T$ only. For ratios use kelvin.
- WATCH OUT:** M in g/mol; $^{\circ}\text{C}$ in T ; confusing energy per molecule with total energy.
- CHECK:** is v_{rms} of order a few hundred m/s at room temperature?

Question

Find the average translational kinetic energy per molecule and the rms speed of (a) nitrogen (N_2 , $M = 28.0 \text{ g/mol}$) and (b) helium ($M = 4.00 \text{ g/mol}$), both at 20°C .

Predict first

Which has the greater kinetic energy per molecule? Which is faster, and by what factor ($\sqrt{28/4}$)?

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Predict first

Which has the greater kinetic energy per molecule? Which is faster, and by what factor ($\sqrt{28/4}$)?

$$T = 293.15 \text{ K.} \quad K = \frac{3}{2}kT = \frac{3}{2}(1.381 \times 10^{-23})(293.15) = 6.07 \times 10^{-21} \text{ J (both gases).}$$

$$v_{\text{rms}}(\text{N}_2) = \sqrt{\frac{3(8.314)(293.15)}{0.0280}} = 511 \text{ m/s;}$$

$$v_{\text{rms}}(\text{He}) = \sqrt{\frac{3(8.314)(293.15)}{0.00400}} = 1.35 \times 10^3 \text{ m/s.}$$

Evaluate: equal K , helium faster by $1352/511 = 2.65 = \sqrt{7}$, as predicted; speeds of order hundreds of m/s, comparable with the speed of sound. ✓

Concept check: two gases at the same temperature

Choose, then explain

Argon ($M = 39.9 \text{ g/mol}$) and neon ($M = 20.2 \text{ g/mol}$) are both at 300 K. How do the rms speed and the average translational kinetic energy per molecule compare?

- A. Argon has the larger K and the larger v_{rms} .
- B. Neon has the larger v_{rms} ; K is the same.
- C. Neon has the larger v_{rms} ; argon has the larger K .
- D. K and v_{rms} are the same for both.
- E. Neon has the larger K ; argon has the larger v_{rms} .

Predict first

Which quantity depends on the mass of the molecule, and which does not?

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- D. K and v_{rms} are the same for both.
- E. Neon has the larger K ; argon has the larger v_{rms} .

Predict first

Which quantity depends on the mass of the molecule, and which does not?

B. $K = \frac{3}{2}kT$ is the same; $v_{\text{rms}} \propto 1/\sqrt{m}$ is larger for the lighter neon, by $\sqrt{39.9/20.2} = 1.41$.

Choose, then explain

10 g of hydrogen (H_2 , $M = 2.02 \text{ g/mol}$) and 10 g of oxygen (O_2 , $M = 32.0 \text{ g/mol}$) are at 300 K. Which sample has the greater *total* translational kinetic energy, and by roughly what factor?

Predict first

Same T , so same energy per molecule. Which sample contains more molecules?

Choose, then explain

10 g of hydrogen (H_2 , $M = 2.02 \text{ g/mol}$) and 10 g of oxygen (O_2 , $M = 32.0 \text{ g/mol}$) are at 300 K. Which sample has the greater *total* translational kinetic energy, and by roughly what factor?

Predict first

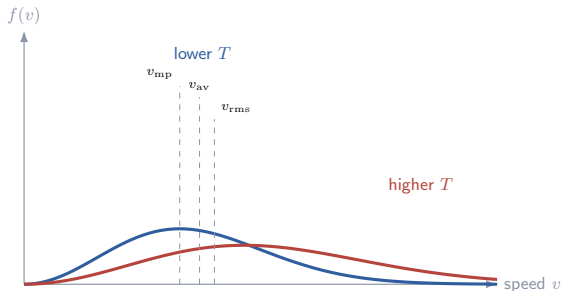
Same T , so same energy per molecule. Which sample contains more molecules?

$$K_{\text{tot}} = \frac{3}{2}nRT = \frac{3}{2}\frac{m}{M}RT.$$

$$\text{H}_2: n = 4.95 \text{ mol}, K_{\text{tot}} = 1.85 \times 10^4 \text{ J}. \quad \text{O}_2: n = 0.3125 \text{ mol}, K_{\text{tot}} = 1.17 \times 10^3 \text{ J}.$$

Evaluate: hydrogen has 15.8 times more, equal to $M_{\text{O}_2}/M_{\text{H}_2}$: same energy per molecule, but 15.8 times as many molecules. ✓

Not all molecules move at the same speed



- The speeds form a **distribution**: a most probable speed, a long high-speed tail.
- Raising T moves the peak to higher speed and widens the curve; the total area (all the molecules) is unchanged.
- Always $v_{mp} < v_{av} < v_{rms}$.

Common error

“All molecules have the speed v_{rms} .” It is only a representative speed; some molecules are much faster and some much slower. Formulas for the distribution are in the supplement.

Your turn: matching speeds

Your turn

- (a) Find the rms speed of carbon dioxide ($M = 44.0 \text{ g/mol}$) at 400 K.
- (b) At what temperature do nitrogen molecules (28.0 g/mol) have the same rms speed as helium molecules (4.00 g/mol) at 300 K?

Predict first

- (b) The nitrogen molecules are heavier. Must the nitrogen be hotter or colder than 300 K? Roughly by what factor?

Your turn: matching speeds

Your turn

- (a) Find the rms speed of carbon dioxide ($M = 44.0 \text{ g/mol}$) at 400 K.
- (b) At what temperature do nitrogen molecules (28.0 g/mol) have the same rms speed as helium molecules (4.00 g/mol) at 300 K?

Predict first

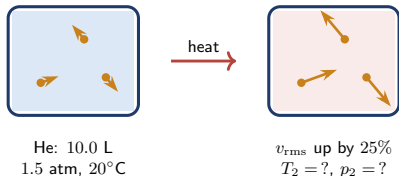
- (b) The nitrogen molecules are heavier. Must the nitrogen be hotter or colder than 300 K? Roughly by what factor?

$$(a) v_{\text{rms}} = \sqrt{\frac{3(8.314)(400)}{0.0440}} = 476 \text{ m/s}.$$

$$(b) \text{ Equal } v_{\text{rms}} \text{ means } T/M \text{ equal: } T_{\text{N}_2} = 300 \times \frac{28.0}{4.00} = 2100 \text{ K}.$$

Evaluate: hotter, by the factor 7 (the ratio of the molar masses), as predicted. ✓

Integrated problem: a heated flask of helium



Decide what physics applies; do not look for a formula

A rigid sealed flask of volume 10.0 L contains helium at 1.50 atm and 20°C. It is heated until the rms speed of the atoms is 25% larger.

- (a) How many moles and atoms are there?
- (b) Find the initial rms speed.
- (c) Find the final temperature and pressure.

Predict first

Sketch the flask before and after. Will T rise by 25%, by more, or by less? Will the pressure rise by the same factor as T ?

Identify, set up: a fixed n in a rigid flask: ideal gas (Toolkit 2) and the kinetic model (Toolkit 7), linked by T . $T_1 = 293.15 \text{ K}$, $p_1 = 1.50 \times 1.013 \times 10^5 = 1.52 \times 10^5 \text{ Pa}$, $V = 10.0 \times 10^{-3} \text{ m}^3$.

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$$(a) \quad n = \frac{p_1 V}{RT_1} = \frac{(1.52 \times 10^5)(10.0 \times 10^{-3})}{(8.314)(293.15)} = 0.623 \text{ mol}; \quad N = nN_A = 3.75 \times 10^{23}.$$

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$$(b) \quad v_{\text{rms},1} = \sqrt{3RT_1/M} = \sqrt{\frac{3(8.314)(293.15)}{4.00 \times 10^{-3}}} = 1.35 \times 10^3\text{ m/s}.$$

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$$(c) \quad v_{\text{rms}} \propto \sqrt{T} \Rightarrow T_2 = (1.25)^2 T_1 = 1.5625 T_1 = 458\text{ K} = 185^\circ\text{C}. \quad V \text{ is constant, so } p_2 = 1.5625 p_1 = 2.34\text{ atm} = 2.37 \times 10^5\text{ Pa}.$$

Evaluate: T rises by 56% (more than 25%), because v_{rms} depends on $\sqrt{T}$; p rises with T by the same factor, as predicted. Check with $pV = nRT$:

$$p_2 = (0.623)(8.314)(458)/(10.0 \times 10^{-3}) = 2.37 \times 10^5\text{ Pa}. \quad \checkmark$$

Extension (unassessed): why the Earth keeps nitrogen but loses hydrogen

In the high atmosphere ($T \approx 1000 \text{ K}$):

$$v_{\text{rms}}(\text{N}_2) \approx 0.94 \text{ km/s}; \quad v_{\text{rms}}(\text{H}_2) \approx 3.5 \text{ km/s}.$$

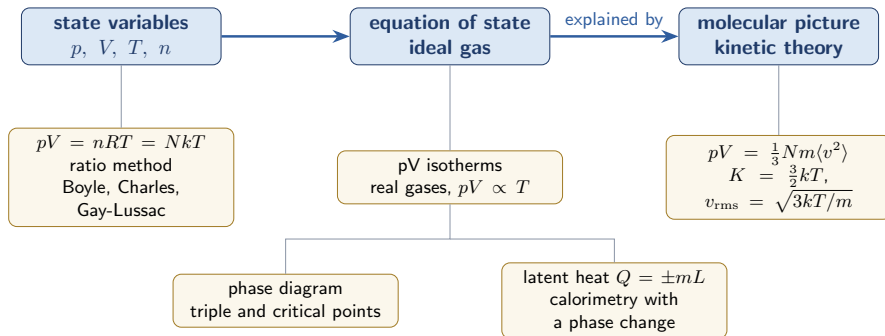
The escape speed from the Earth is 11.2 km/s (given). The ratio $v_{\text{esc}}/v_{\text{rms}}$ is about 12 for N_2 and about 3.2 for H_2 .

The tail matters

A rule of thumb: a gas is kept for billions of years only if v_{esc} exceeds about $6v_{\text{rms}}$. For hydrogen, enough molecules in the high-speed tail exceed v_{esc} and escape.

An application of the speed distribution; it is not assessed. The escape speed is derived in the Gravitation lecture.

Lecture 10 concept map



MASTER PROCEDURE – THERMAL PROPERTIES

1. **IDENTIFY** The system, its phase and the process. Is the amount of gas fixed?

2. **SKETCH** The system, the path on the pV or phase diagram, or the molecule and wall.

3. **UNITS** Kelvin, absolute pressure, m^3 , moles, M in kg/mol .

4. **PREDICT** The direction of change, a limiting case and the rough size.

5. **SELECT THE MODEL** Equation of state, ratio method, diagram, latent heat, calorimetry or kinetic theory.

6. **EXECUTE** Write the equations first, then substitute with units.

7. **EVALUATE** Units, signs, magnitude, limiting cases, and a second check.

SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2

SET UP $\leftrightarrow$ steps 3–5

EXECUTE $\leftrightarrow$ step 6

EVALUATE $\leftrightarrow$ step 7

Physical meaning

Next lecture: the work done by a gas as its volume changes (the area under a path on the pV diagram) and the first law of thermodynamics.



THE SEVEN TOOLKITS

- | | | |
|---|--|---|
| 1 | State variables and units | kelvin; absolute pressure; m^3 ; moles |
| 2 | The ideal-gas equation | $pV = nRT = NkT$; $n = m/M$ |
| 3 | The ratio method | $p_1 V_1 / T_1 = p_2 V_2 / T_2$; what is constant? |
| 4 | Reading pV and phase diagrams | $T \propto pV$; paths; triple and critical points |
| 5 | Heating curves and latent heat | $mc\Delta T$ in a phase; $\pm mL$ at a change |
| 6 | Calorimetry with a phase change | decide the final state first; $\sum Q = 0$ |
| 7 | Kinetic theory | $K = \frac{3}{2}kT$; $v_{\text{rms}} = \sqrt{3kT/m}$; scaling |

The Master Procedure sits above these. Earlier toolkits (units, energy, signs, $Q = mc\Delta T$) are reused, not renamed.

Amount

$$n = m/M, \quad N = nN_A, \quad m_0 = M/N_A$$

Ideal gas

$$pV = nRT = NkT, \quad \rho = pM/(RT)$$

Change of state (n fixed)

$$p_1V_1/T_1 = p_2V_2/T_2$$

Latent heat

$$Q = \pm mL, \quad Q = mc\Delta T \text{ within a phase}$$

Pressure (kinetic)

$$pV = \frac{1}{3}Nm(v^2)_{\text{av}}$$

Energy and temperature

$$K = \frac{1}{2}m(v^2)_{\text{av}} = \frac{3}{2}kT, \quad K_{\text{tot}} = \frac{3}{2}nRT$$

Speed

$$v_{\text{rms}} = \sqrt{3kT/m} = \sqrt{3RT/M}$$

Constants

$$R = 8.314 \text{ J/(mol K)},$$

$$k = 1.381 \times 10^{-23} \text{ J/K},$$

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

Six habits for Lecture 10

✓ Habit 1

Convert first: kelvin, absolute pressure, SI volume.

✓ Habit 2

Ask what is fixed: the amount, T , p or V .

✓ Habit 3

Sketch the isotherm or the path before you compare states.

✓ Habit 4

Decide the final state before you write $\sum Q = 0$ with a phase change.

✓ Habit 5

Separate energy per molecule from total energy.

✓ Habit 6

Use ratios and scaling ($\propto \sqrt{T}$, $1/\sqrt{M}$) to check a result.

Checklist: can I ...

- ☐ define the state variables and use n , N , M ?
- ☐ convert to kelvin, absolute pressure and SI volume?
- ☐ use $pV = nRT$ and $pV = NkT$?
- ☐ use the ratio method and relate it to Boyle, Charles and Gay-Lussac?
- ☐ compare states on a pV diagram?
- ☐ read a phase diagram, including the triple and critical points?
- ☐ calculate $Q = \pm mL$ and a heating curve?
- ☐ solve calorimetry with a change of phase?
- ☐ state the assumptions of the kinetic model?
- ☐ explain pressure from molecular collisions?
- ☐ use $K = \frac{3}{2}kT$ and $v_{\text{rms}} = \sqrt{3kT/m}$?
- ☐ compare two gases at the same temperature?
- ☐ combine the gas law with the kinetic model?
- ☐ select the appropriate model in an unfamiliar problem?

After this week

Tutorial sheet: extra gas-law (including gauge pressure), pV , heating-curve, phase-change calorimetry, kinetic-theory and combined problems. Supplementary notes: the ideal gas as a model, the kinetic derivation, the Maxwell speed distribution, mean free path, equipartition (preview), real gases and the water phase diagram.